Potential Target Region Extraction and Isolated Forest With Statistical Correlation Representation for Hyperspectral Anomaly Detection

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***Abstract*— Anomaly detection is a hot topic in the field of hyperspectral image (HSI) processing, which aims to find anomalous targets through spatial and spectral differences from a complex background. However, there still remains two unsolved problems in anomaly detection methods: 1) since the global image covers more complex scenes, it inevitably suffers a lot of false alarms in the anomaly detection and 2) the sensitivity of the target to the global mode or the local mode detector is different. A detector that only considers one mode usually misses some targets, which makes the detection rate low. This article proposes an anomaly detection method for HSIs with potential target region extraction. The potential target region is extracted through the combination of the tensor robust principal component analysis (RPCA) and the curvature circle model. The obtained subregions are used in the region traversing method to filter out the areas not containing the target. In addition, in order to adapt to the different targets contained in the region, the proposed method combines the isolated forest of the global model and the statistical correlation of the local model, thereby enhancing the separability of the target and the background. Experimental results show that the proposed method can effectively extract potential target regions and greatly reduce the false alarm rate. Furthermore, the proposed method can highlight the target for a higher detection rate and accuracy.**

***Index Terms*— Anomaly detection, global and local modes, hyperspectral image (HSI), isolated forest, potential target region extraction, statistical correlation.**

I. Introduction

HYPERSPECTRAL imaging provides rich spectral information for the materials in a scene, which allows distinguishing different ground objects precisely [1]–[4]. Accordingly, it has been widely used in various applications: land cover classification [5], military reconnaissance [6], mineral exploration [7], and target/anomaly detection [8].

Especially, hyperspectral target detection is considered one of the most important remote sensing tasks due to its significance in many military and civilian applications. The goal of target detection is to localize target pixels in the scene from complex backgrounds, which can be formulated as a binary classification problem. It is also formulated as anomaly target detection if the prior knowledge is not given or is challenging to obtain. Since obtaining the prior spectral information of the target is challenging in practical applications, anomaly detection is considered an essential task in various applications such as mineral detection, civil search and rescue, environmental monitoring, and food and drug quality detection [9]. However, anomaly detection still remains an unsolved challenging problem due to the lack of prior spectral information for anomaly targets [8].

Over the past several decades, many anomaly detection methods have been studied. The most classic anomaly detection method is Reed–Xiaoli (RX) method [10]. The RX detectors can be categorized into two variants according to the background description: the local RX (LRX) and the global RX (GRX) [11]. Neighboring pixels are considered the background in the LRX, while the entire image is considered the background in the GRX. The RX method measures the Mahalanobis distance between the pixel and backgrounds to detect anomalies under the assumption of Gaussian multivariate normal distribution. However, the application of the RX method is limited due to noise, inaccurate background modeling, and computational complexity. Many works were conducted to improve the RX detector, including the kernel RX (KRX)-method [12], the subspace-based RX detector [13], and the cluster KRX (CKRX) method [14].

Since the background distribution of hyperspectral images (HSIs) is more complicated, the Gaussian distribution assumption is often invalid. In order to overcome this problem, many other approaches for anomaly detection methods were explored. The local summation anomaly detection (LSAD) method [15] was proposed by utilizing the spatial and spectral joint information and characteristics, requiring no background distribution consideration. Deep learning-based methods were also explored for HSI anomaly detection, taking advantage of extracting the high-level features [16]. In [17], the autoencoder network was adopted to reconstruct the background and to compare the original and decoded images for anomaly detection. Li *et al.* [18] introduced transfer learning to train the convolutional neural network for HSI anomaly detection. The blocked adaptive computationally efficient outlier nominator (BACON) [19] optimizes the background model by

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using robust statistics estimated from the entire HSI scene subsets.

In recent years, the low-rank and sparse decomposition theory has been successfully used in hyperspectral anomaly detection, where the hyperspectral imagery is decomposed into low-rank and sparse components containing anomalous targets [20]. The anomalies and noise were separately modeled in the low-rank and sparse matrix decomposition (LRaSMD) [21]. However, the weak anomalies were not distinguished from the noise. In the low-rank and sparse decomposition model with the mixture of Gaussian (LSDM-MoG) model [22], the mixture Gaussian was used to model the sparse component to describe the complex noise containing anomaly. A non-Gaussian detector was also designed based on the Manhattan distance. The robust principal component analysis (RPCA) was used to achieve LRaSMD [23], and then the spare component was detected by the RX algorithm, called the RPCA-RX.

More recently, representation-based detectors have been introduced into anomaly detection, which can detect targets without any assumptions on background distribution and make full use of spatial information [24]. Li and Du [25] proposed a collaborative-representation-based detector (CRD) based on the concept that background pixels can be approximately represented by their neighboring pixels, unlike the pixels of the anomaly. Imani [26] extended the CRD based on morphology (MCRD) to exploit the spatial information better. Wang *et al.* [27] proposed a self-weighted collaborative representation-based detector (SWCRD). Cheng and Wang [28] proposed a graph and total variation regularized low-rank representation (GTVLRR) model, considering the potential spatial information between the target pixel and its neighboring pixels. In this method, the representation coefficient was regularized by the graph Laplacian, and the local manifold structure was preserved with the increased computational complexity. Many other outstanding anomaly detectors have also been proposed. The background joint sparse representation detector (BJSRD) [29] adopted the joint sparse representation to detect anomalies, where the matched projection length of the target pixel on the orthogonal complementary background subspace was measured. Tao *et al.* [30] proposed the fractional Fourier transform-based RX (FrFE-RX) for increased noise suppression and discrimination capability of the detector.

However, most anomaly detectors were limited to certain conditions due to the background modeling scheme [31]. Furthermore, there exist unsolved problems that limit the practical application of anomaly detection.

First, the target includes detailed information since it belongs to the high-frequency information in the image. Thus, many false alarms can occur in the detection of global images. The existing anomaly detectors use a manually cropped HSI from an entire scene image, which does not satisfy practical application requirements. The false detection rate is significantly increased when the global image with a complex scene is used in most existing detection methods [32]. Therefore, how to automatically extract potential target regions is an essential problem for anomaly detection. Second, most of the anomaly detection methods only consider the global situation, such as GRX and LRaSMD, or only consider local neighborhoods, such as LRX and CRD. Since different abnormal targets have different sensitivity to global detectors and local detectors, only one mode could lead to missed targets, resulting in a lower detection rate. Consequently, properly combining the global and local detection modes is important to improve the detection rate.

This article proposes a potential target region extraction and isolated forest with statistical correlation representation (PRE-IS) for anomaly detection of the global HSIs containing the complex scene. The proposed method mainly consists of the potential target region extraction and anomaly detection processes. Usually, the target exists in the background with low-rank characteristics, and the target distribution has sparse characteristics. The potential anomaly extraction is to treat the HSI as a third-order tensor. Through the tensor RPCA model with full variational regularization, it can be divided into a low-rank tensor and sparse tensor, thus obtaining low-rank feature maps and sparse feature maps. Then, the curvature circle model for the low-rank feature maps is used to determine the boundaries of the subregions that exhibit low-rank characteristics in the image. Finally, the region containing the target is obtained by determining the potential target area of the traversal area twice. The anomaly detection process is based on the extracted potential target area. The detection result is obtained from a global perspective through the isolation forest method. The statistical correlation is used to express the pixels to be measured through the statistical feature pairs of the local area, determining the potential abnormal targets. The final detection result is combined with the above two results by introducing an adjustment factor. The overall diagram of the proposed method is shown in Fig. 1.

The main contributions of this article can be summarized as follows.

1) A novel potential target region extraction method is proposed, named region of interest extraction. The sparse feature of low-rank backgrounds that include sparse targets is obtained through a tensor RPCA model for the global HSI containing the complex scene. The curvature circle model and two traversal processes are adopted to remove regions that do not contain the target, thereby reducing the false alarm rate of anomaly detection. To the best of our knowledge, this article is the pioneering attempt to extract potential target regions for the global HSI with a complex scene in anomaly detection.
2) The statistical correlation expression of the isolated forest of the global and local models is used for each potential target area. Also, the adjustment factor is adopted to effectively combine the global and local results, enhancing the abnormal target in the potential target area. In such a way, the proposed method can be applied to various targets with an improved detection rate.

The rest of this article is organized as follows. Section II briefly introduces the related works. Section III provides

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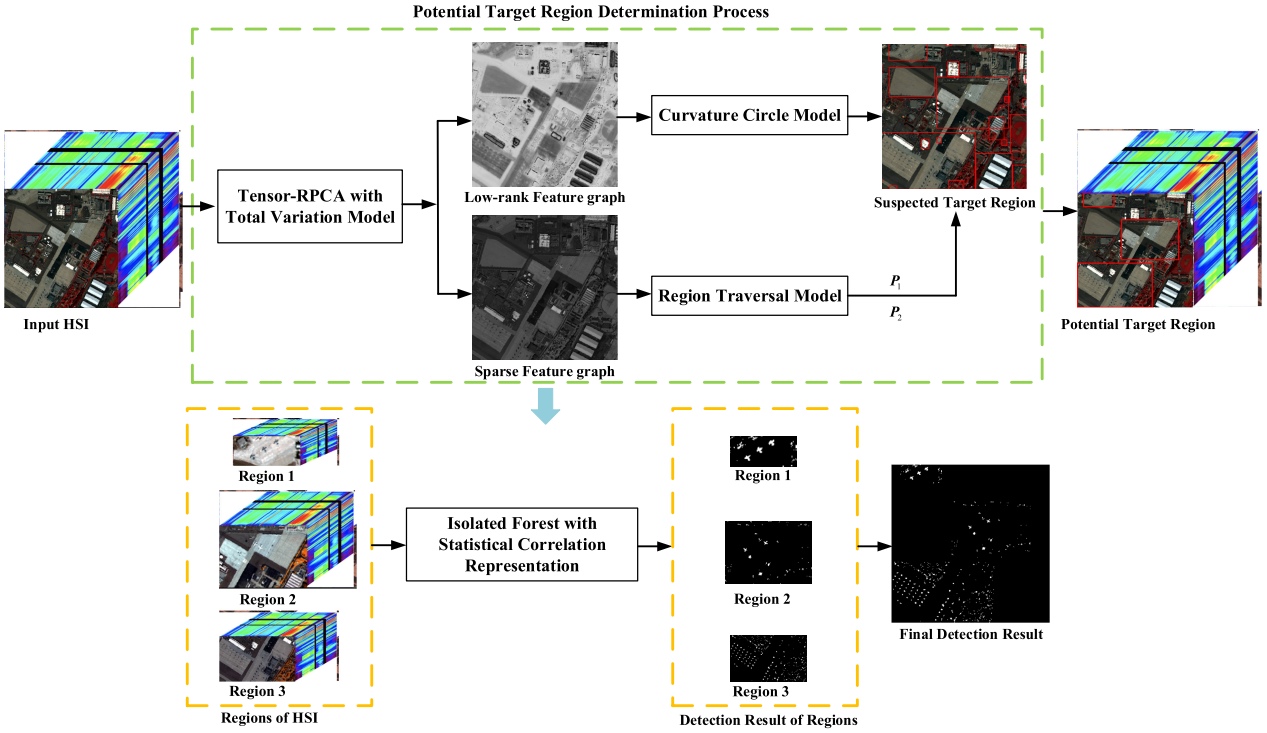


Fig. 1. Overview of the proposed PRE-IS method for anomaly detection.

a detailed description of the proposed method. Section IV presents the experimental data and analysis results. Finally, conclusions are drawn in Section V.

II. Related Works

A. Total Variation Regularization and Linear TV-Norm

The total variation regularization (TV) can enhance the piecewise smoothness in spatial dimensions while maintaining the sharp edges between the different regions [33]. The TV norm is defined as l_1-norm by

$$\mathrm{TV}(\mathbf{X}) = \sum_{\{i,j\}\in\varepsilon} \|\mathbf{x}_i - \mathbf{x}_j\|_1 \tag{1}$$

where $\mathbf{X}\in R^{c\times w\times n}$ is a HSI, c and w represent the spatial dimensions, and n represents the spectral dimension. $\mathbf{x}_i$ and $\mathbf{x}_j$ are spatially adjacent neighboring pixel pairs (in the horizontal and vertical directions), and ε is the spatial pixel set of $\mathbf{X}$.

The energy function of the TV-norm is minimized by the gradient method to obtain the solution since no corresponding closed-form solution exists. For efficient computation, while preserving the edge information and promoting piecewise smoothness, the linear TV (LTV)-norm is proposed [34] instead of the TV-norm. It is defined as the F-norm of the difference between the neighbor pixels

$$\mathrm{LTV}(\mathbf{X}) = \|\mathbf{H}_\mathrm{h}\mathbf{X}\|_\mathrm{F}^2 + \|\mathbf{H}_\mathrm{v}\mathbf{X}\|_\mathrm{F}^2 \tag{2}$$

where $\mathbf{H}_\mathrm{h}$ is the linear operator computing the difference of representation coefficients between each pixel and its horizontal neighboring pixels, formulated as $\mathbf{H}_\mathrm{h}\mathbf{X} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_{c-1}]$. $\mathbf{h}_i = \mathbf{x}_i - \mathbf{x}_{i\mathrm{h}}$ represents the difference between the ith pixel and its horizontal neighboring pixels. The linear operator $\mathbf{H}_\mathrm{v}$ is similar to $\mathbf{H}_\mathrm{h}$, which computes the vertical differences, and its definition is $\mathbf{H}_\mathrm{v}\mathbf{X} = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{w-1}]$; $\mathbf{v}_i = \mathbf{v}_i - \mathbf{v}_{i\mathrm{h}}$ also represents the difference between the ith pixel and its neighboring vertical pixel. $\|\cdot\|_F^2$ is computed by

$$\|\mathbf{A}\|_F^2 = \sum_{i,j} |\mathbf{a}_{i,j}|^2 \tag{3}$$

where $\mathbf{A}$ is a matrix and $\mathbf{a}_{i,j}$ is the element in $\mathbf{A}$.

B. Tensor Notations

In this section, the notations of tensor would be introduced and given the tensor representation of hyperspectral imagery data [35], [36]. This article follows the relevant provisions of linear algebra. The bold upper-case letters represent matrices, such as $\mathbf{X}$ and $\mathbf{Y}$, while the lower-case letters represent vectors, such as $\mathbf{x}$. Note that nonbold letters represent scalars. The Euler script letters are applied to represent tensors, e.g., Ξ.

The HSI data have three dimensions which are a typical third-order tensor. In the tensor theory, the matrix obtained by fixing one dimension unchanged but keeping two dimensions change is called the slice of a third-order tensor. The vector obtained by fixing two dimensions unchanged but keeping the one dimension change is called the fiber of a third-order tensor. For an HSI tensor $\Xi \in R^{c\times w\times n}$, $\Xi(k,:,:)$, $\Xi(:,k,:)$, and $\Xi(:,:,k)$ denote the horizontal, lateral, and frontal slices, respectively. The frontal slices are the image matrixes of all bands in HSI representing the two spatial dimensions. $\Xi(:,i,j)$, $\Xi(i,:,j)$, and $\Xi(i,j,:)$ denote the mode-1, mode-2, and mode-3 fibers, respectively. The mode-3 fiber is the spectral vector of the ground objects in HSI containing the

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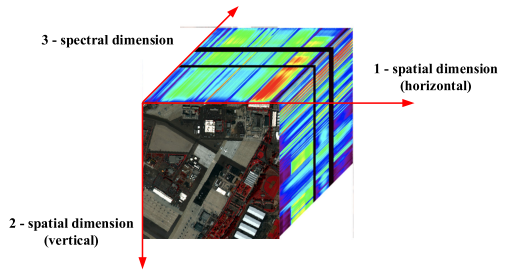


Fig. 2. Correlation interpretation of HSI tensor.

pixels representing the one spectral dimension information, as shown in Fig. 2. The mode-i unfolding is the operation of tensor matricization, where the tensor Ξ is the matrix denoted by $\mathbf{X}_{(i)}$ that is obtained by arranging the mode-i fibers as the columns of the matrix and denoted by $\mathbf{X}_{(i)} = \text{unfold}_i(\Xi)$ and $\Xi = \text{fold}_i(\mathbf{X}_{(i)})$. For example, the HSI tensor X can be unfolded along the first mode ($i = 1$), which is defined as $\text{unfold}_1(\Xi) = \mathbf{X}_{(1)} \in R^{c\times(wn)}$ or $\Xi = \text{fold}_1(\mathbf{X}_{(1)})$.

III. Proposed Method

A. Tensor-RPCA Model With Total Variation

As seen in [37], the HSI tensor is usually decomposed into a low-rank tensor and a sparse tensor for anomaly detection, which represent the background and anomaly targets in a HSI, respectively. Generally, the low-rank property is constrained by the nuclear norm regularization, and the sparse property is constrained by $l_{2,1}$-norm regularization. In global HSIs, anomaly targets also exist in low-rank background regions, whose distribution is sparse in these regions. Therefore, it is crucial to find the boundary of regional property change from the low-rank tensor. The boundary is reflected in the different ground objects represented by the spatial pixels. In order to highlight the region boundary while maintaining the spatial relationship within the image, the TV regularization constraint is introduced into the tensor decomposition process. The cost function of the LTV-norm regularized tensor decomposition can be written as follows:

$$\begin{aligned} &\min_{\mathcal{L},\mathcal{S}} \|L_3\|_* + \lambda\|S_3\|_{2,1} + \tau\mathbf{LTV}(\mathcal{L}) \\ &\text{s.t. } \mathcal{Y} = \mathcal{L} + \mathcal{S} \end{aligned} \tag{4}$$

where $\Psi \in R^{c\times w\times n}$ is HSI tensor, Λ is the low-rank tensor, Σ is sparse tensor, $\mathbf{L}_{(i)} = \text{unfold}_i(\Lambda)$ (i = 1,2,3), and $\mathbf{S}_{(3)} = \text{unfold}_i(\Sigma)$.

λ and τ are tradeoff parameters. $\|\cdot\|_*$ denotes nuclear norm of the $\mathbf{L}_3$, computed by summation of the singular values [20]. $\|\cdot\|_{2,1}$ is the $l_{2,1}$ norm and is computed as

$$\|\mathbf{X}\|_{2,1} = \sum_{j=1}^{c\times\mathbf{w}} \sqrt{\sum_{i=1}^{n} (\mathbf{X}_{i,j})^2} \tag{5}$$

and

$$\text{LTV}(\mathcal{L}) = \|\mathbf{H}_h\mathbf{L}_{(1)}\|_F^2 + \|\mathbf{H}_v\mathbf{L}_{(2)}\|_F^2 \tag{6}$$

where $\mathbf{L}_{(1)} \in R^{c\times(wn)}$ represents the matrix that unfolds in the horizontal direction of spatial dimension, and $\mathbf{L}_{(2)} \in R^{w\times(cn)}$ represents the matrix that unfolds in the vertical direction. $\mathbf{H}_\text{h}$ is the linear horizontal operator such that $\mathbf{H}_\text{h}\mathbf{X} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_{c-1}] \in R^{(c-1)\times wn}$, and $\mathbf{h}_i = \mathbf{l}_{\mathbf{1}(i)} - \mathbf{l}_{1(i-1)}$ represents the difference between horizontal direction pixel spectral vectors. The vertical linear operator $\mathbf{H}_\text{v}$ is similar as $\mathbf{H}_\text{h}$, defined as $\mathbf{H}_\text{v}\mathbf{X} = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{w-1}] \in R^{(c-1)\times wn}$. $\mathbf{v}_j = \mathbf{l}_{2(j)} - \mathbf{l}_{2(j-1)}$ also represents the difference between the ith pixel and its neighboring vertical pixel.

In order to solve the problem (4), it is converted into the following equivalent equations by introducing auxiliary variables $\mathbf{V}_1$, $\mathbf{V}_2$, and $\mathbf{V}_3$ to make the variables separable:

$$\begin{aligned} &\min_{\mathcal{L},\mathcal{S}} \|V_3\|_* + \lambda\|S_3\|_{2,1} + \tau\left(\|H_hV_1\|_F^2 + \|H_vV_2\|_F^2\right) \\ &\text{s.t.}\mathcal{Y} = \mathcal{L} + \mathcal{S}, V_1 = L_1, V_2 = L_2, V_3 = L_3. \end{aligned} \tag{7}$$

The augmented Lagrangian function of (7) is

$$\begin{aligned} &L(\mathcal{L},\mathcal{S},\mathbf{V}_1,\mathbf{V}_2,\mathbf{V}_3,\mathbf{T}_1,\mathbf{T}_2,\mathbf{T}_3,\mathbf{T}_4) \\ &= \|\mathbf{V}_3\|_* + \lambda\|\mathbf{S}_{(3)}\|_{2,1} + \tau\left(\|\mathbf{H}_h\mathbf{V}_1\|_F^2 + \|\mathbf{H}_v\mathbf{V}_2\|_F^2\right) \\ &\quad + \frac{\mu}{2}\|\mathbf{L}_{(3)} - \mathbf{V}_3 + \mathbf{T}_3\|_F^2 + \frac{\mu}{2}\|\mathbf{L}_{(1)} - \mathbf{V}_1 + \mathbf{T}_1\|_F^2 \\ &\quad + \|\mathbf{L}_{(2)} - \mathbf{V}_2 + \mathbf{T}_2\|_F^2 + \frac{\mu}{2}\|\mathbf{Y}_{(3)} - \mathbf{L}_{(3)} - \mathbf{S}_{(3)} - \mathbf{T}_4\|_F^2\Big) \end{aligned} \tag{8}$$

where $\mathbf{T}_1$, $\mathbf{T}_2$, $\mathbf{T}_3$, and $\mathbf{T}_4$ are Lagrange multipliers, and $\mu > 0$ is penalty parameters. The variables are alternately updated while fixing other variables in the direction of minimizing the augmented Lagrangian function L. The detailed procedure of solving (8) with alternating direction method of multipliers (ADMM) [38] is as follows.

1) $\mathbf{V}_1$ is updated, while other variables are fixed. The optimization can be solved as

$$\begin{aligned} \mathbf{V}_1^{k+1} &= \arg\min_{\mathbf{V}_1} \tau\|\mathbf{H}_h\mathbf{V}_1^k\|_F^2 + \frac{\mu}{2}\left(\|\mathbf{L}_{(1)}^k - \mathbf{V}_1^k + \mathbf{T}_1^k\|_F^2\right) \\ &= \frac{\mu}{2\tau}\left(\mathbf{H}_h^T\mathbf{H}_h + \mu\mathbf{I}_h\right)^{-1}\mathbf{L}_{(1)}^k + \mathbf{T}_1^k \end{aligned} \tag{9}$$

where k is the index of iteration, and $\mathbf{I}_h \in^{h\times h}$ is the identity matrix.

2) $\mathbf{V}_2$ is updated, while other variables are fixed through the following optimization problem:

$$\begin{aligned} \mathbf{V}_2^{k+1} &= \arg\min_{\mathbf{V}_2} \tau\|\mathbf{H}_v\mathbf{V}_2^k\|_F^2 + \frac{\mu}{2}\left(\|\mathbf{L}_{(2)}^k - \mathbf{V}_2^k + \mathbf{T}_2^k\|_F^2\right) \\ &= \frac{\mu}{2\tau}\left(\mathbf{H}_v^T\mathbf{H}_v + \mu\mathbf{I}_v\right)^{-1}\mathbf{L}_{(2)}^k + \mathbf{T}_2^k. \end{aligned} \tag{10}$$

3) $\mathbf{V}_3$ is updated, while other variables are fixed through the following optimization problem:

$$\mathbf{V}_3^{k+1} = \arg\min_{\mathbf{V}_3} \|\mathbf{V}_3^k\|_* + \frac{\mu}{2}\left(\|\mathbf{L}_{(3)}^k - \mathbf{V}_3^k + \mathbf{T}_3^k\|_F^2\right). \tag{11}$$

The problem (11) is solved by singular value thresholding operator [39]

$$\mathbf{V}_3^{k+1} = \mathcal{D}_{1/\mu}\left(\mathbf{L}_{(3)}^k + \mathbf{T}_3^k\right) \tag{12}$$

where $\mathcal{D}$ is the singular value thresholding operator.

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4) Λ is updated, while other variables are fixed through the following optimization:

$$\mathcal{L}^{k+1} = \frac{1}{4}\left[\sum_{i=1}^{3} \text{fold}_i\left(\mathbf{V}_i^{k+1} - \mathbf{T}_i^k\right) + \left(\mathcal{Y} - \mathcal{S}^k\right)\right] - \frac{1}{4\mu}\text{fold}_3\left(\mathbf{Y}_4^k\right). \tag{13}$$

The updated results of $\mathbf{L}_{(i)}$ $(i = 1,2,3)$ are as follows:

$$\mathbf{L}_{(i)}^{k+1} = \text{unfold}_i\left(\mathcal{L}^{k+1}\right). \tag{14}$$

5) Σ and $\mathbf{S}_{(3)}$ are updated, while other variables are fixed through the following optimization problem:

$$\mathbf{S}_{(3)}^{k+1} = \arg\min_{\mathcal{S}_3} \lambda\left\|\mathbf{S}_{(3)}^k\right\|_{2,1} + \frac{\mu}{2}\left\|\mathbf{Y}_{(3)} - \mathbf{L}_3^k - \mathbf{S}_{(3)}^k + \mathbf{T}_4^k\right\|_F^2. \tag{15}$$

The problem (15) can be solved by the $l_{2,1}$-minimization operator [40], obtaining a closed-form solution defined as

$$\mathbf{S}_{(3)}^{k+1} = \Omega_{\lambda/\mu}\left(\mathbf{Y}_{(3)} - \mathbf{L}_3^k + \mathbf{T}_4^k\right) \tag{16}$$

where Ω is the $l_{2,1}$-minimization operator, and as a result, Σ is updated as

$$\mathcal{S} = \text{fold}_3\left(\mathbf{S}_{(3)}\right). \tag{17}$$

6) The Lagrange multipliers $\mathbf{Y}_1$, $\mathbf{Y}_2$, $\mathbf{Y}_3$, and $\mathbf{Y}_4$ are updated with the following formula:

$$\mathbf{T}_1^{k+1} = \mathbf{T}_1^k + \mathbf{L}_{(1)}^{k+1} - \mathbf{V}_1^{k+1} \tag{18}$$
$$\mathbf{T}_2^{k+1} = \mathbf{T}_2^k + \mathbf{L}_{(2)}^{k+1} - \mathbf{V}_2^{k+1} \tag{19}$$
$$\mathbf{T}_3^{k+1} = \mathbf{T}_3^k + \mathbf{L}_{(3)}^{k+1} - \mathbf{V}_3^{k+1} \tag{20}$$
$$\mathbf{T}_4^{k+1} = \mathbf{T}_4^k - \left(\mathbf{Y}_{(3)} - \mathbf{L}_{(3)}^{k+1} - \mathbf{S}_{(3)}^{k+1}\right). \tag{21}$$

The iterative process is terminated when the stopping criteria (22) are satisfied or reached maximum iterations $K_{\max}$. The tensor-RPCA with the TV process is summarized in Algorithm 1

$$\left\|\mathcal{Y}_3 - \mathcal{L}_3^{k+1} - \mathcal{S}_3^{k+1}\right\|_2^F \le \varepsilon. \tag{22}$$

After satisfying the optimal termination condition, the solutions of Λ and Σ can be obtained, and the corresponding low-rank feature graph $\boldsymbol{l}$ and sparse feature graph $\boldsymbol{s}$ can be calculated as follows:

$$\mathbf{l}(i,j) = \sqrt{\sum_{k=1}^{n} |\mathcal{L}(i,j,k)|^2} \tag{23}$$

$$\mathbf{s}(i,j) = \sqrt{\sum_{k=1}^{n} |\mathcal{S}(i,j,k)|^2}. \tag{24}$$

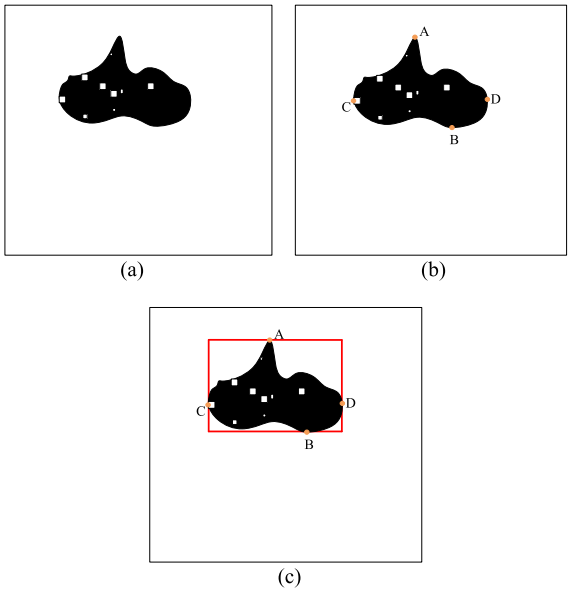


Fig. 3. Potential target area determination: (a) abstract low-rank feature graph; (b) regional boundary points; and (c) rectangle identifies the potential target area.

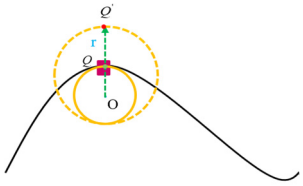


Fig. 4. Detailed process of curvature circle model.

B. Curvature Circle Model for Region Boundary Determination

The obtained low-rank feature map is used to find the background low-rank area. Then, the suspected target area is further refined using the sparse feature map. The low-rank feature map is converted into a binary image via the OTSU method [41] to highlight regions with low-rank characteristics. Then, the smallest rectangular containing the potential target area is estimated, as shown in Fig. 3.

At this time, the suspected target could be located exactly on the boundary, like the vertex C in Fig. 3(b). In this case, the target could be divided when the boundary is determined in Fig. 3(c), affecting the anomaly detection accuracy.

In order to prevent this problem, this article proposes a curvature circle model to determine the suspected target area, as shown in Fig. 4. Suppose that the center of curvature circle that is orthogonal to the boundary is O, and the suspected target in Fig. 4 is located at the boundary point Q. Then, along the direction of the ray OQ, the point is extended to Q' with the parameter r. Q is used as the new boundary point to determine the target area enclosed by the rectangular box with Q' so that the boundary points have been expanded. The parameter r is determined by the actual size of the target and the spatial resolution of the image. The detailed analysis of the parameter r is elaborated in Section IV-C.

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Algorithm 1 The Tensor-RPCA With Total Variation Model

Input: Hyperspectral tensor $\mathcal{Y} \in R^{c\times w\times n}$, parameters $\lambda > 0$, $\tau > 0$, $\mu > 0$ and K_{max}.

Initialize: $\Lambda = \Sigma = 0$, $\mathbf{V}_1 = \mathbf{V}_2 = \mathbf{V}_3 = 0$, $\mathbf{T}_1 = \mathbf{T}_2 = \mathbf{T}_3 = \mathbf{T}_4 = 0$, $\mathbf{L}_{(i)} = \text{unfold}_i(\Lambda)$, $\mathbf{S}_{(3)} = \text{unfold}_3(\Sigma)$, $\tau = 0.2$, $\mu = 10^{-3}$, $\mu_{max} = 10^{10}$, $\rho = 1.5$, $\varepsilon = 10^{-4}$ and $K_{max} = 500$.

While not converged, **do**

1. Fix others and update $\mathbf{V}_1$, $\mathbf{V}_2$, $\mathbf{V}_3$ with (9), (10), and (12), respectively.
2. Fix others and update Λ by (13), and update $\mathbf{L}_{(i)}$ (i = 1,2,3) by (14).
4. Fix others and update $\mathbf{S}_{(3)}$ by (16) and update Σ by (17).
5. Update $\mathbf{T}_1$, $\mathbf{T}_2$, $\mathbf{T}_3$, and $\mathbf{T}_4$ by (18)-(21).
6. Update μ by $\mu = \min(\rho\mu, \mu_{max})$.
7. Check the convergence condition by (22).
8. k = k + 1.

End while

9. Calculate the low-rank feature graph l and the sparse feature graph s by (23)-(24).

Output: the optimal solution Λ, Σ, and the low-rank feature graph $\boldsymbol{l}$, the sparse feature graph $\boldsymbol{s}$.

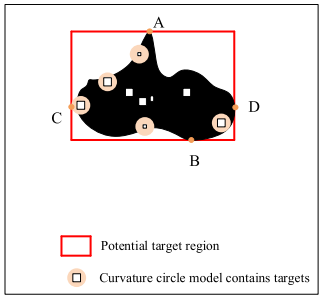


Fig. 5. Determination of potential target area by curvature circle model.

Fig. 5 shows an example of the final potential target area determined by the curvature circle model. The white rectangle represents the target, and the orange circle represents the circle of curvature located at the boundary of the region. Through the proposed curvature circle model, the left vertex C of the potential target area (red rectangle) can completely contain the target pixel. Note that the target located in the boundary region is included in the rectangular box, which is beneficial to the subsequent detection process.

C. Potential Target Region Determination Process

The process of determining the potential target area is mainly to analyze the characteristics related to the target in each area obtained from Section II-B, including the sparse characteristics in the area and the high-frequency information in the area, combined with the environment to get the final result. For the potential target area, the determination process includes the process of traversing the area twice. The traversal area refers to traversing the entire area from the front to the back with a specific step, as shown in Fig. 6. The first traversal is based on the maximum possible length of the target, and the second traversal is based on the minimum possible length of the target. The existence of a target is determined through the two area traverse. The core idea is motivated by the squeeze theorem of the limit theory in advanced mathematics. When the maximum and minimum possible lengths of the target are estimated appropriately, they apply to different targets in the image.

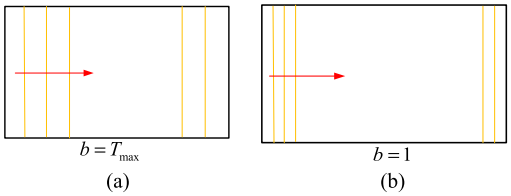


Fig. 6. Schematic of traversing the region when: (a) stride is T_{max} and (b) stride is 1.

For the target region, the increase of the sparsity is usually induced by the targets. This sparsity for each region is measured based on sparse feature maps, computed as follows. First, the sparse feature map is transformed into a binary image $\mathbf{B}$ by the OTSU method. The sparsity for each area is computed by the ratio of the nonzero pixels in the area to the total number of pixels in $\mathbf{B}$

$$K = \frac{\text{count}(\text{ROI}_i > 1)}{N_i} \tag{25}$$

where ROI_i represents the suspected target area, N_i represents the total number of all pixels in the area, and $\text{count}(\text{ROI}_i > 1)$ indicates the number of nonzero pixels in the binary image corresponding to the area. For the low-rank feature that contains the background in the area, the area corresponding to a low ratio of the rank feature map to the area size is expressed.

Because the sparse feature map reflects the target information to a certain extent, it has a kind of high-frequency information. The Shannon information entropy is used to measure the amount of high-frequency information, defined as

$$E = -\sum_{i=0}^{L-1} h_i \log(h_i) \tag{26}$$

where L represents the range of gray levels of $\boldsymbol{X}$, and h_i is the probability that the ith gray level occurs in the image. h_i is computed by mapping the sparse feature map $\mathbf{s}$ gray value range of the region to [0, 255]. Then, the number of times each gray value i appears is divided by the total number of pixels in s to obtain the value of h_i.

The first traversal is based on the maximum length of the target. Through one side of the fixed area, from the other side of the area, the maximum possible length of the target is used as the traversal stride T_{max} to traverse the area from front to back on the other side of the area, as shown in Fig. 6(a). The index P_1 obtained by the first traversal to measure whether the area contains the target is

$$P_1 = \text{score}\left[\frac{K * \left(\sum_{i=1}^{U} E_i\right)}{R/(m \times n)}\right] \begin{cases} \geq T_1 \\ < T_1 \end{cases} \tag{27}$$

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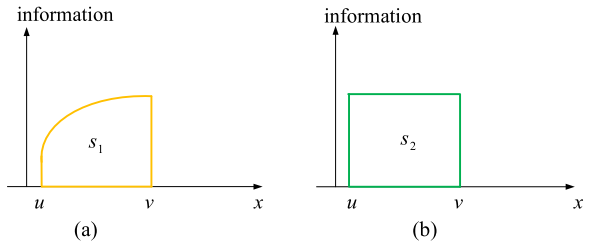


Fig. 7. Schematic of information entropy curve: (a) s_1 and (b) s_2.

where E_i represents the information entropy of the region at traversal step i, R represents the rank of the feature map of the region, and U represents the total number of traversal steps. m and n are the length and width of the region, respectively, and (27) reflects the extent to which the low-rank background contains sparse target information. With the movement of the traversal stride, the lower the rank of the feature map, the sparser the feature map, and the greater the amount of high-frequency information. Also, it indicates a greater possibility of targets in this region.

The judgment threshold T_1 is determined according to the environment where the image is located. The specific setting is discussed in detail in Section IV-C. A higher P_1 value than T_1 indicates that the background of this region shows low-rank characteristics and the suspected targets show sparse characteristics, indicating that more likely the region contains targets.

The second traversal process is the main consideration of the growing degree of high-frequency information in the sparse feature map. Taking the traversal stride as 1, one of the sides is fixed to traverse the area, as shown in Fig. 6. When the area is traversed, the information entropy corresponding to each step is calculated. After the traversal computation, the corresponding information entropy forms a curve. The area s_1 surrounded by the curve and the coordinate axis and the rectangular area s_2 determined by the coordinates of the curve is calculated, as shown in Fig. 7.

s_1 is the area enclosed by the curve and $x = u$, $x = v$, $y = 0$, and s_2 is the area within the rectangular area determined by the vertex of the curve and $x = u$, $x = v$, $y = 0$. The ratio P_2 of s_1 and s_2 is used to measure the increase of regional high-frequency information

$$P_2 = \frac{s_1}{s_2} \begin{cases} \geq T_2 \\ < T_2 \end{cases} \tag{28}$$

where the judgment threshold T_2 is also related to the environment in which the image is located. The specific setting is also discussed in Section IV-C. A higher P_2 value of the area than T_2 indicates that the area contains the target.

When the value of P_2 is less than T_2, the high-frequency information is decreasing which means that the low-frequency information increases in the traversal processes. Thus, it can be considered that no target exists in this area.

D. Isolated Forest With Statistical Correlation for Anomaly Detection

The anomalies are detected based on the potential target regions. Then, the detection result graphs of each region are obtained and combined according to the spatial regions in the original HSI.

The anomaly detection for the potential target regions is conducted in two processes. First, the isolated forest method is used to obtain the anomaly detection result $\mathbf{r}_{\text{if}}$ through the sparse part of the corresponding region. Then, the statistical correlation representation (SCR) method is used to detect the corresponding anomaly through the original image part of the corresponding region. Finally, those two results are combined to obtain the final detection results. The specific flow diagram of anomaly detection process is shown in Fig. 8.

The concept of the term isolation from other objects was first proposed by Liu *et al.* [42], where a collection of binary trees was adopted to randomly partition instances until all of them were isolated. The anomalies that participate in a few propagations of the image are apt to have shorter paths in the binary tree. Thus, the anomalies having more distinguishable attributes can be more easily separated [43]. Through this partitioning process, the anomalies commonly have a shorter path than the normal instances. This binary tree is called isolation trees (iTrees). Since discrete points representing the easily isolated target pixels, unlike other points representing background pixels, induce a shallower depth on the iTree, they are more likely to be anomalies when a forest of iTrees (an iForest) collectively produces shorter path lengths for some points [44]. The swamping and masking effects are reduced by the iForest model through a repeated subsample collection. Such an isolation concept can be applied for hyperspectral anomaly detection due to the rarity and distinct characteristics of the anomalies in the image scene. Furthermore, the sparse part in the potential abnormal region of the image has already been obtained by the tensor decomposition model, and thus the anomalies can be easily distinguished from the background pixels by iForest.

1) Growth Process of Isolation Forest: First, the potential target area is denoted by $\mathbf{Y}^*$, and $\mathbf{Y}^* \in R^{c*w*\times n}$. The M pixels are selected from $\mathbf{Y}^*$ and divided into two child nodes: left node $\mathbf{Y}_{\text{l}}$ and right node $\mathbf{Y}_{\text{r}}$. M is set to 300 in this article. The node partition selects a pixel in M, and q is the number of bands randomly selected, ranging in [1, n]. ω represents the set of pixel thresholds (ω_1, ω_2, and ...) of $\mathbf{Y}^*$ on the qth band, which randomly determines the node (left or right) that the current pixel belongs to. $\mathbf{Y}_q^m$ represents the value of the qth band of the mth pixel of Y^*. If $\mathbf{Y}_q^m$ in $\mathbf{Y}^*$ is less than ω, the mth pixel is categorized into the left node; if $\mathbf{Y}_q^m$ is greater than ω, the mth pixel is categorized into the right node. Then, according to the node partition criteria, the pixels in the two child nodes $\mathbf{Y}_{\text{l}}$ and $\mathbf{Y}_{\text{r}}$ can be further divided iteratively until the termination condition is satisfied. The growth process of the iTree is depicted in Fig. 9. The iteration termination conditions are as follows.

1) The height of the binary tree reaches the maximum value $H_{\max}$.
2) The number of pixels in each child node is 1, or the value of pixels in each subnode is the same.

The maximum height $H_{\max}$ is determined as follows to maximize the algorithm efficiency:

$$H_{\max} = \log_2(M). \tag{29}$$

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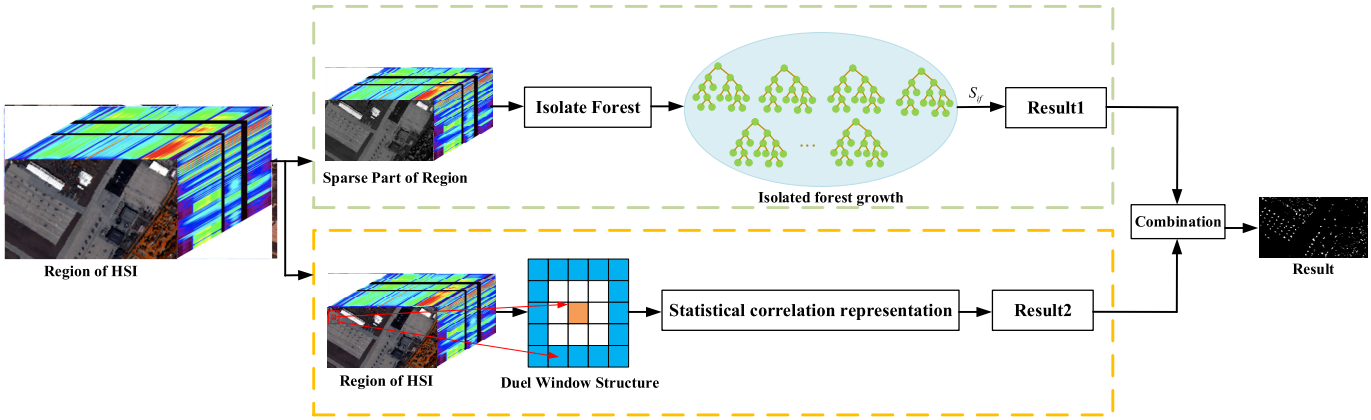


Fig. 8. Flow diagram of anomaly detection.

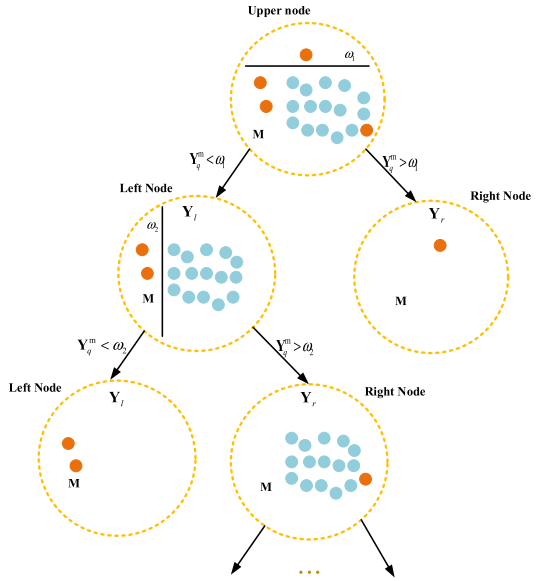


Fig. 9. Graphical analysis of the growth process of isolated tree.

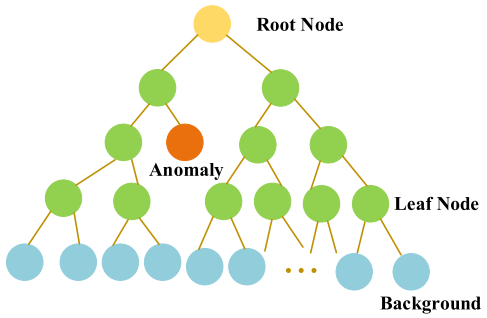


Fig. 10. iTree structure diagram.

As shown in Fig. 10, the iTree consists of binary trees of randomly selected samples, where each node represents a single HSI pixel or multiple HSI pixels with similar spectral values. Due to the majority of background pixels over the abnormal pixels, most of the selected sample pixels belong to the background. The background is usually located at the bottom of the iTree since the background pixels are difficult to be isolated due to their higher frequency. Because the amount of abnormal pixels is small, there is a certain difference between them and the background pixels. Thus, it is relatively easy to isolate the abnormal pixels, as intuitively described in Fig. 9.

Multiple binary trees are constructed to form a random forest by continuously randomly selecting different pixel sets from $\mathbf{Y}^*$. The growth process of the binary tree is repeated t times to obtain an isolated forest. Here, t is set to 1000; thereby, the number of trees in the isolated forest is 1000, ensuring that all pixels in the region $\mathbf{Y}^*$ are allocated to the binary tree of the isolation forest.

2) Anomaly Scores Computation Using Isolating Forest: The average path length of the constructed iTrees in the forest is used to determine the anomalies. The path length $e(x_i)$ is defined as the distance from the root node to the edge node of the pixel in the iTree T_i $(i = 1{:}\ t)$ contained in the isolated forest. The average path length $E(e(x_i))$ of the pixel in the isolated forest is the average of the path length of the pixel to each iTree. Due to the minority of abnormal pixels, the corresponding average path length for the anomalies is shorter than background pixels. Accordingly, the average path length can be used to detect anomalies [45]. The anomaly score is computed based on the average path, defined as follows:

$$S_{if} = 2^{-\frac{E(e(x_i))}{p(M)}} \tag{30}$$

where M is the size of randomly selected subsamples, and $p(M)$ represents the average height of the tree defined as

$$p(M) = 2H(M-1) - (2(M-1)/M) \tag{31}$$

where $H(\cdot)$ is a harmonic number being approximately estimated by $\ln(\cdot) + 0.5772156649$ (Euler constant). As a result, the anomaly detection result $\mathbf{r}_{\text{if}}$ is obtained from the random forest.

As the next step, the SCR is constructed. The neighboring pixel matrix $\mathbf{Y}_n = \{\mathbf{y}^j\}_{j=1}^n$ indicates the pixels between the

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outer and inner windows, where $\mathbf{y}$ is a test pixel from $\mathbf{Y}^*$, and z is the number of pixels in $\mathbf{Y}_n$. When $\mathbf{y}$ is the background, it can be fully represented by using neighboring pixels $\mathbf{Y}_n$; otherwise, it cannot. The two hypotheses can be obtained: the null hypothesis H_0 indicates the absence of the anomaly and the alternative hypothesis H_1 indicates the existence of the anomaly

$$H_0 : \mathbf{y} = \mathbf{Y}_n\sigma \text{ (Anomalous target absent)} \tag{32}$$

$$H_1 : \mathbf{y} \neq \mathbf{Y}_n\sigma \text{ (Anomalous target present)} \tag{33}$$

where σ is a weight vector. Inspired by the collaborative representation detector [25], the SCR distinguishes the pixel to be measured from the background by introducing statistical features. The corresponding Lagrange function is defined as

$$l = \arg\min \|\mathbf{y} - \mathbf{Y}_n\boldsymbol{\sigma}\|_F^2 + \lambda\|\boldsymbol{\sigma}\|_F^2. \tag{34}$$

The following correlation matrix $\mathbf{C}_y$ is used for increased correlation constraint [46]:

$$\mathbf{C}_y = \begin{pmatrix} \|\mathbf{y} - \mathbf{x}_1\|_2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \|\mathbf{y} - \mathbf{x}_s\|_2 \end{pmatrix}. \tag{35}$$

Then, the Lagrange function is rewritten as

$$l_2 = \arg\min \left\|\mathbf{y} - \mathbf{Y}_n\boldsymbol{\sigma}\right\|_F^2 + \lambda\left\|\mathbf{C}_y\boldsymbol{\sigma}\right\|_F^2. \tag{36}$$

Taking the extremum of σ, we obtain the following equation:

$$\boldsymbol{\sigma} = \left(\mathbf{Y}_n^T\mathbf{Y}_n + \lambda\mathbf{C}_y^T\mathbf{C}_y\right)^{-1}\mathbf{Y}_n^T\mathbf{y}. \tag{37}$$

The background representation in the pixel $\mathbf{y}^*$ to be measured is $\mathbf{Y}_n\boldsymbol{\sigma}$, and the detection criterion based on the constant false-alarm rate of the test pixel is defined by

$$\mathbf{r}_{\text{sc}} = (\mathbf{y} - \mathbf{Y}_n\boldsymbol{\sigma})^T\boldsymbol{\Gamma}^{-1}(\mathbf{y} - \mathbf{Y}_n\boldsymbol{\sigma}) \tag{38}$$

where Γ is the background covariance matrix. In this way, the anomaly is detected with SCR: $\mathbf{r}_{\text{sc}}$.

Finally, the anomaly of the potential target region is detected by combining $\mathbf{r}_{\text{if}}$ and $\mathbf{r}_{\text{sc}}$

$$\mathbf{r} = \mathbf{r}_{\text{if}} + \varphi\mathbf{r}_{\text{sc}} \tag{39}$$

where φ is the parameter, which is related to the relative values of $\mathbf{r}_{\text{if}}$ and $\mathbf{r}_{\text{sc}}$. The optimal parameters are empirically obtained and discussed in Section IV-C. The process of proposed method is summarized in Algorithm 2.

Algorithm 2 The Proposed PRE-IS Method for Anomaly Detection

Inputs: HSI, Parameter r, Parameter T1 and T2, Optimal window size settings and adjustment factor φ.

Steps:

1. Obtaining the low-rank feature map and sparse feature map through the tensor-RPCA with total variation by **Algorithm 1**.
2. According to the low-rank feature map, the low-rank regions are obtained through the curvature circle model, and the potential target regions are obtained by the potential target region determination process.
3. Obtaining global mode detection result $\mathbf{r}_{if}$ through the isolated forest with (30), respectively.
4. Obtaining local mode detection result $\mathbf{r}_{sc}$ through statistical correlation representation with (38) respectively.
4. Calculating the final detection result by combining $\mathbf{r}_{if}$ and $\mathbf{r}_{sc}$ with appropriate regulator φ by (39).

IV. Experimental Results

A. Hyperspectral Image Dataset

In order to verify the effectiveness of the proposed method, the following HSI datasets were used.

1) San Diego Dataset (HSI-I): This HSI dataset was obtained in the San Diego airport area, CA, USA, with the AVIRIS sensor [33]. Each band contains 400 × 400 pixels, where the low SNR, water absorption regions, and bad bands were removed. The targets in this dataset are 47 helicopters and six fighters.

2) Gulfport Dataset (HSI-II): This dataset was collected with the AVRIS sensor in the Gulfport airport area, CA. Each band contains 300 × 300 pixels in each band, where the noisy bands were removed. The targets in this dataset are four helicopters and three fighters.

3) San Francisco Port Dataset (HSI-V and HSI-VI): This dataset was collected with the AVIRIS sensor in the San Francisco Bay area port. Two HSIs are included, where the noisy bands were removed. The targets in this dataset are ships.

4) Avon Dataset (HSI-V): This dataset was collected with a push-broom hyperspectral ProSpec TIR-VS sensor during the "SpecTIR Hyperspectral Airborne Experiment 2012" (SHARE 2012) data collection campaign [47]. A region of 330 × 330 pixels covering Avon, South of Rochester, NY, USA, was used for our experiments. This dataset includes 360 bands with a spectral range of 400–2450 nm (a spectral resolution of 5 nm). The detection targets include three red or blue felts, 25 tarps with 68 target pixels in this dataset [24].

Fig. 11 shows the examples of these HSI and the corresponding anomaly targets. Table I summarizes the detailed information of these datasets.

B. Evaluation Criteria and Compared Methods

The robustness and effectiveness of the proposed method are evaluated compared with the other HSI anomaly detection methods. The compared methods include GRX [10], LRX [11], LSDM-MoG [22], RPCA-RX [23], LSAD [15], GTVLRR [28], and LRaSMD [21], which are selected according to the contribution and the representativeness. The LRX and LSAD are the local mode detectors, while the GRX, LSDM-MoG, RPCA-RX, GTVLRR, and LRaSMD are the global mode detectors. As the most commonly used method, the global pattern detector is suitable for simple-scene images and has low computational complexity and higher efficiency. A partial mode detector is suitable for images with more complex but small-scale scenes. It takes more computational complexity than global model detectors.

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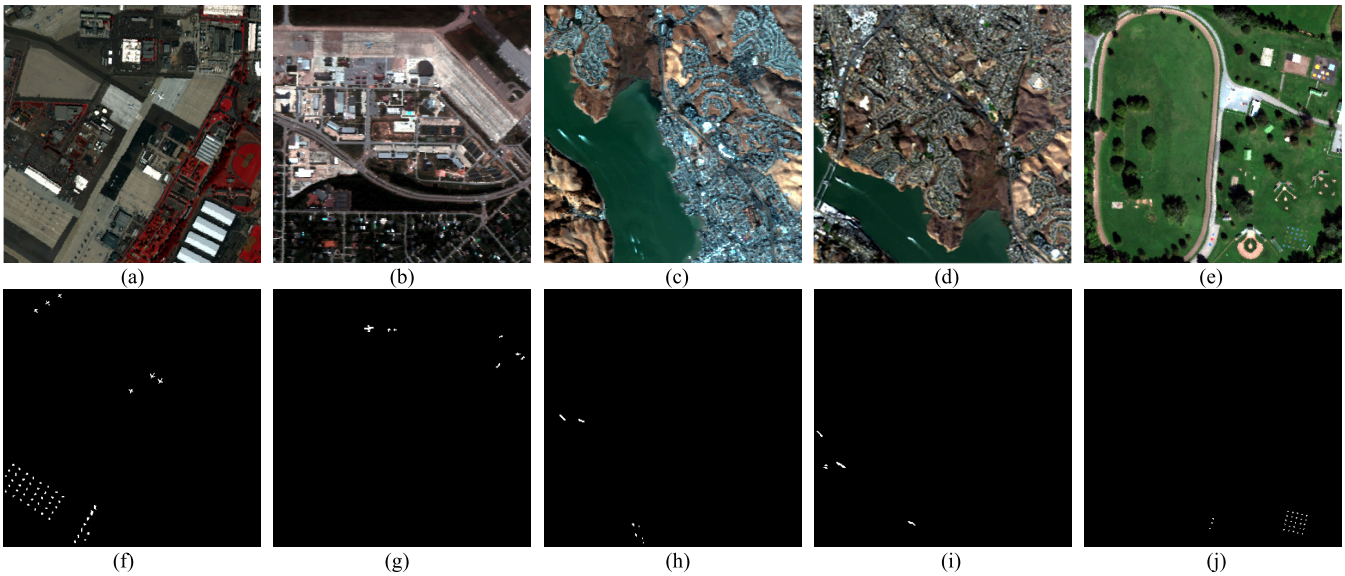


Fig. 11. Examples of HSI datasets I and V: (a)–(e) false-color images of HSI-I and HSI-V and (f)–(j) corresponding ground truths for (a)–(e).

TABLE I
DETAIL INFORMATION OF THESE SIX HYPERSPECTRAL IMAGES

HSIs	Sizes	Captured location	Spatial resolution	Spectral range	Number of bands	Sensors
HSI-I	400×400	San Diego	3.5m	0.37μm-2.51μm	186	AVIRIS
HSI -II	300×300	Gulfport	3.4m	0.36μm-2.49μm	193	AVIRIS
HSI -III	300×300	San Francisco	16.4m	0.36μm-2.49μm	194	AVIRIS
HSI -IV	300×300	San Francisco	16.4m	0.36μm-2.49μm	191	AVIRIS
HSI -V	330×330	Rochester	1m	0.40μm-2.45μm	360	ProSpec

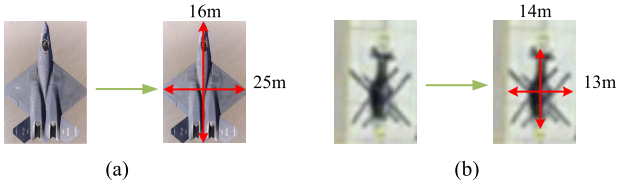


Fig. 12. Real size estimation of aircraft: (a) fighter and (b) helicopter.

The receiver operating characteristic (ROC) curve and area under the ROC curve (AUC) value are used as quantitative evaluation indices [48]. The detection probability P_D and the false alarm rate P_{FA} are inevitably tradeoffs since a low threshold gives a high P_D, while a high threshold gives a low P_{FA}. The ROC curve can describe the tradeoff between those two performance matrices under the Neyman–Person criterion. A detector with high P_D while maintaining P_{FA} is typically considered the better one. A larger value of AUC indicates better detection performance.

C. Parameter Analysis

In this section, the influence of parameters on anomaly detection performance is discussed. Three parameters r, T_1, and T_2 are included in the process of extracting potential abnormal regions. In the anomaly detection process, two parameters are involved: the window size and the parameter φ.

1) Parameter r: Since each HSI has a different spatial resolution and targets, the parameter r affects the potential target region extraction process. The scene of HSI-I and HSI-II is the airport, which contains the targets of fighter and helicopter, and the spatial resolution is similar, and thus they can use the same parameter r. HSI-III and HSI-IV are port scenes, and targets are ships, and thus they can also use the same parameter r.

Hyperspectral anomaly detection is based on the application target as the traction, such as aircraft and ships. The parameter r is determined according to the proportion of the target in the image, which is affected by the target size and the image spectral resolution. The target size $a \times b$ is divided by the image spatial resolution G to obtain the radius r^* of the circumscribed circle that can be closely connected to the target. The parameter r is set as greater than r^*, as follows:

$$r^* = \max\left(a/G, b/G\right) \tag{40}$$

$$r > \lceil r^* \rceil. \tag{41}$$

An aircraft is taken as an example to analyze the obtaining process of the parameter r. The targets included in HSI-1 are fighter planes and helicopters. Usually, the size of a fighter plane is 25 m × 16 m, and the size of a helicopter is about 14 m × 13 m, as shown in Fig. 12. As shown in Table I, the spatial resolution of HSI-I is 3.5 m. Thus, the size of the image occupied by the fighter is about 8 × 5, and the radius r^* that can completely cover the circumcircle of the fighter should be about 4. In the same way, according to the real size of the helicopter, the size in the HSI can be obtained to be approximately 4 × 2, and the radius r^* of the circumcircle of the helicopter should be approximately equal to 2 to completely cover the helicopters. Therefore, in order to be able to completely cover the aircraft targets, and considering the direction of the aircraft parked to the airport,

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TABLE II
OPTIMAL CIRCUMSCRIBED CIRCLE RADIUS ON THE DIFFERENT HSIS

	HSI-I	HSI-II	HSI-III	HSI-IV	HSI-V
r^*	4	4	6	6	1
r	6	6	8	8	2
T_{max}	8	8	12	12	2

and the inevitable stretching during the imaging process, the optimal r of the aircraft is determined to be 6, as shown in Fig. 13. Since the spatial resolution of HSI-II is similar to that of HSI-I, r can also be equally set to 6.

According to the above analysis idea, when the targets contained in the HSI are ships, the best parameter setting r of HSI-III and HSI-IV can be obtained as 8. The HSI-V contains the colored canvas covering the entire pixel of the target. Similarly, it can be obtained that r is equal to 2. Table II summarizes the r^* and r values of all experimental HSIs.

2) Parameters T_1 and T_2: In the potential target region extraction, this article proposes the area traversing method. T_1 is the threshold value of the index P_1 of the first traversal area when the stride is $T_{\max}$, and T_2 is the threshold value of the index P_2 of the second convenient area when the stride is 1.

Parameters T_1 and T_2 are determined according to the scene. First, $T_{\max}$ should be determined according to the experimental image. In the process of analyzing the parameter r, the optimal circumscribed circle radius r^* that can closely cover the target is obtained, and thus $T_{\max}$ of each image is twice that of r^*

$$T_{\max} = 2r^*. \tag{42}$$

Because the index P_1 determines whether the sparsity enhancement and high-frequency information increase due to the entry of the target in the local low-rank area, due to the difficulty in mathematical computation, an approximated value is obtained via experimental methods. This article uses the thirteen data of the airport, beach, and city from the ABU dataset [49] to estimate the index P_1. In order to avoid the influence of the target orientation and shadow, data augmentation with rotations was adopted. The original data were rotated with the angles of 0°, 45°, 90°, 135°, 180°, 270°, and 315°. In total, 104 augmented data were used. Fig. 14 shows the examples of the expanded-ABU dataset.

Similarly, the index P_2 determines the degree of growth of high-frequency information in the region, and the extended ABU dataset is also used to estimate the approximate value of the index P_2.

The expanded-ABU dataset is taken as a region sample known to contain anomaly targets, and the corresponding index P_1 and P_2 values are calculated. The P_1 and P_2 values of the images in the corresponding dataset are formed into a graph, as shown in Fig. 15.

Through the curve, we can get the P_1 and P_2 values that can represent the airport, port, and city (the red line in Fig. 15 indicates the mean value of the data), and thus we can take the corresponding P_1 and P_2 values as the T_1 and T_2 in the potential target region determination. The T_1 and T_2

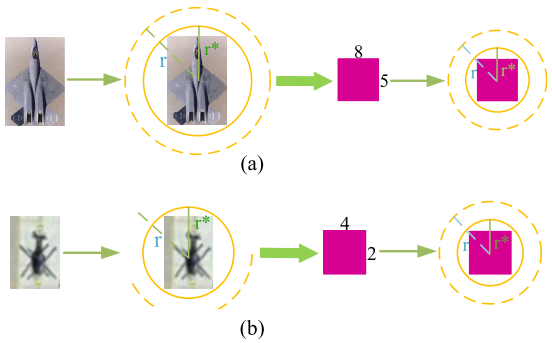


Fig. 13. Schematic of determining parameter r in HSI-I: (a) fighter and (b) helicopter.

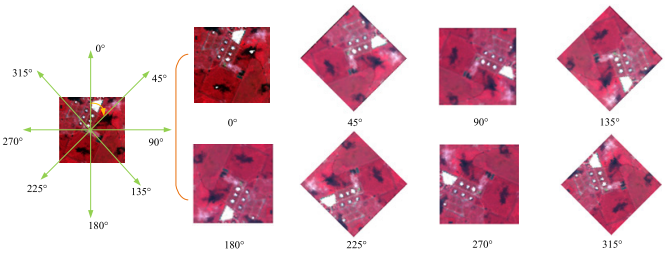


Fig. 14. Schematic of rotation expansion of indicator verification dataset.

TABLE III
OPTIMAL PARAMETERS T_1 AND T_2 SETTING ON THE DIFFERENT HSIS

	HSI-I	HSI-II	HSI-III	HSI-IV	HSI-V
T_1	39.82	39.82	1011.90	1011.90	158.85
T_2	0.9368	0.9368	0.8175	0.8175	0.9291

TABLE IV
OPTIMAL DOUBLE WINDOW SETTING ON THE REGIONS IN DIFFERENT HSIS

	HSI-I	HSI-II	HSI-III	HSI-IV	HSI-V
Region 1	(21,17)	(21,17)	(17,15)	(21,19)	(13,7)
Region 2	(21,17)	(21,17)	—	—	—
Region 3	(13,7)	—	—	—	—

values of each scene image in the experiment are summarized in Table III.

3) Window Size and the Parameter φ: The window size and parameter φ are the influencing factors in the anomaly detection process, which is determined by experiments according to the targets contained in the area in this article. The window parameters are determined to achieve the highest AUC value, as shown in Table IV. The AUC value of the detection result is analyzed in detail in Section IV-D.

The parameter φ is the adjustment factor of the detection result, which is related to the values of $\mathbf{r}_{\text{if}}$ and $\mathbf{r}_{\text{sc}}$. As the adjustment factor, it is also determined by maximizing the AUC value of the detection result, as shown in Table V.

It can be seen that the value of different regulatory factors φ has a certain impact on the detection results. When the AUC value of test results reaches the maximum, the corresponding regulatory factor φ value is the best regulatory

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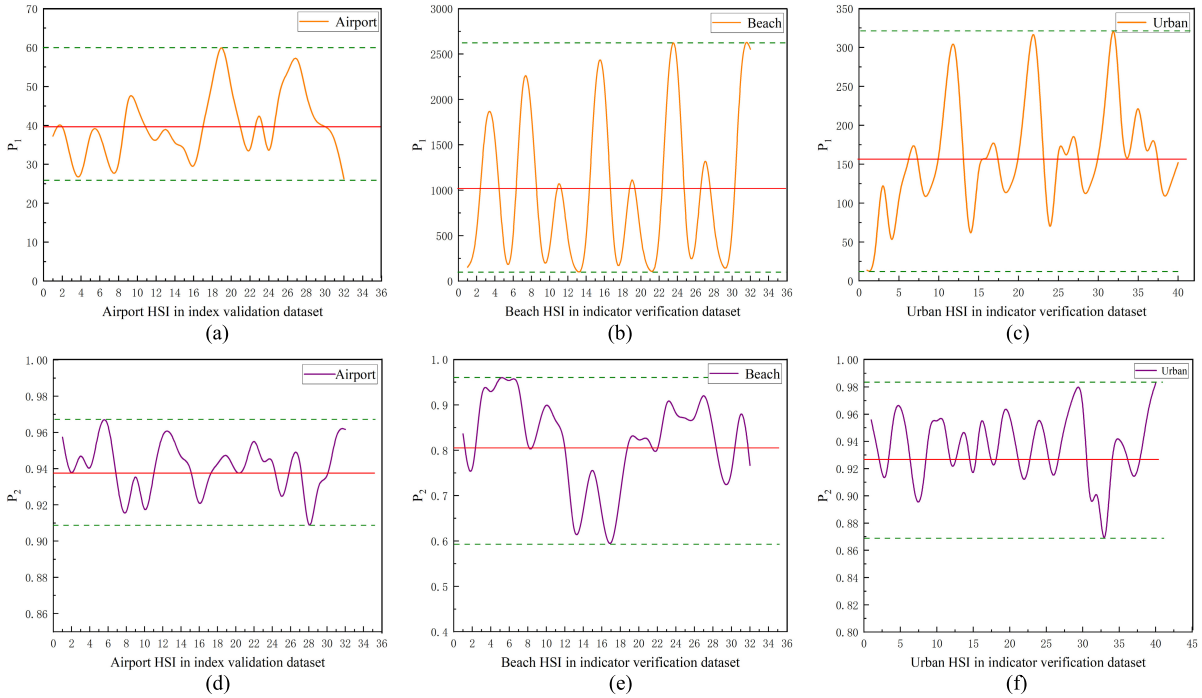


Fig. 15. Curve formed by P_1 and P_2 values of the index validation dataset: (a) P_1 curve for the airport scene; (b) P_1 curve for the beach scene; (c) P_1 curve for the urban scene; (d) P_2 curve for the airport scene; (e) P_2 curve for the beach scene; and (f) P_2 curve for the urban scene.



Fig. 16. Potential target region determination results. (a) HSI-I. (b) HSI-II. (c) HSI-III. (d) HSI-IV. (e) HSI-V.

TABLE V
AUC METRICS ON THE DIFFERENT HSIS BY THE PARAMETER Φ

	10^{-3}	10^{-2}	10^{-1}	10^{0}	10^{1}	10^{2}	10^{3}
HSI-I	**0.9889**	0.9829	0.9229	0.8549	0.8405	0.8389	0.8387
HSI-II	**0.9855**	0.9849	0.9829	0.9724	0.9657	0.9642	0.9641
HSI-III	0.9743	0.9782	0.9805	0.9865	**0.9881**	0.9879	0.9846
HSI-IV	0.9537	0.9642	0.9895	0.9895	0.9897	0.9902	**0.9926**
HSI-V	0.9970	**0.9979**	0.9969	0.9942	0.9921	0.9897	0.9767

factor. AUC values corresponding to the best regulatory factors are shown in bold in Table V.

As indicated in the analysis, in practical applications, the indicators T_1 and T_2 are obtained from the ABU dataset that has been rotated and expanded. The dual-window parameters for anomaly detection can be obtained based on the radius of the circumscribed circle of the target. Thus, the practicability of the proposed method is proven. The parameters r and φ are needed to be determined, which is also caused by the uncertainty of the target area in the global image and the increased complexity of the background grounds.

D. Detection Performance

This section compares the proposed method with seven existing anomaly detection models: GRX, LRX, LSDM-MoG, RPCA-RX LSAD, GTVLRR, and LRaSMD. First, potential target region extraction is compared, as shown in Fig. 16. As shown in Fig. 16, the proposed method can effectively find the potential region containing the targets. The tensor decomposition process can highlight the low-rank sparse characteristics of the target region, and the curvature circle model can protect it well. The integrity of the target accurately determines the potential target area through the two-step traversal process of the selected area, which is beneficial to the anomaly detection process for the global HSI with complex scenes.

Then, the anomaly detection on five HSI datasets is qualitatively compared in Fig. 17. Note that for each dataset, optimal parameters were set. The graphs show that the proposed method accurately detects the area by combining the isolated forest and the local SCR method for each area. It can achieve better and robust detection results for different scene

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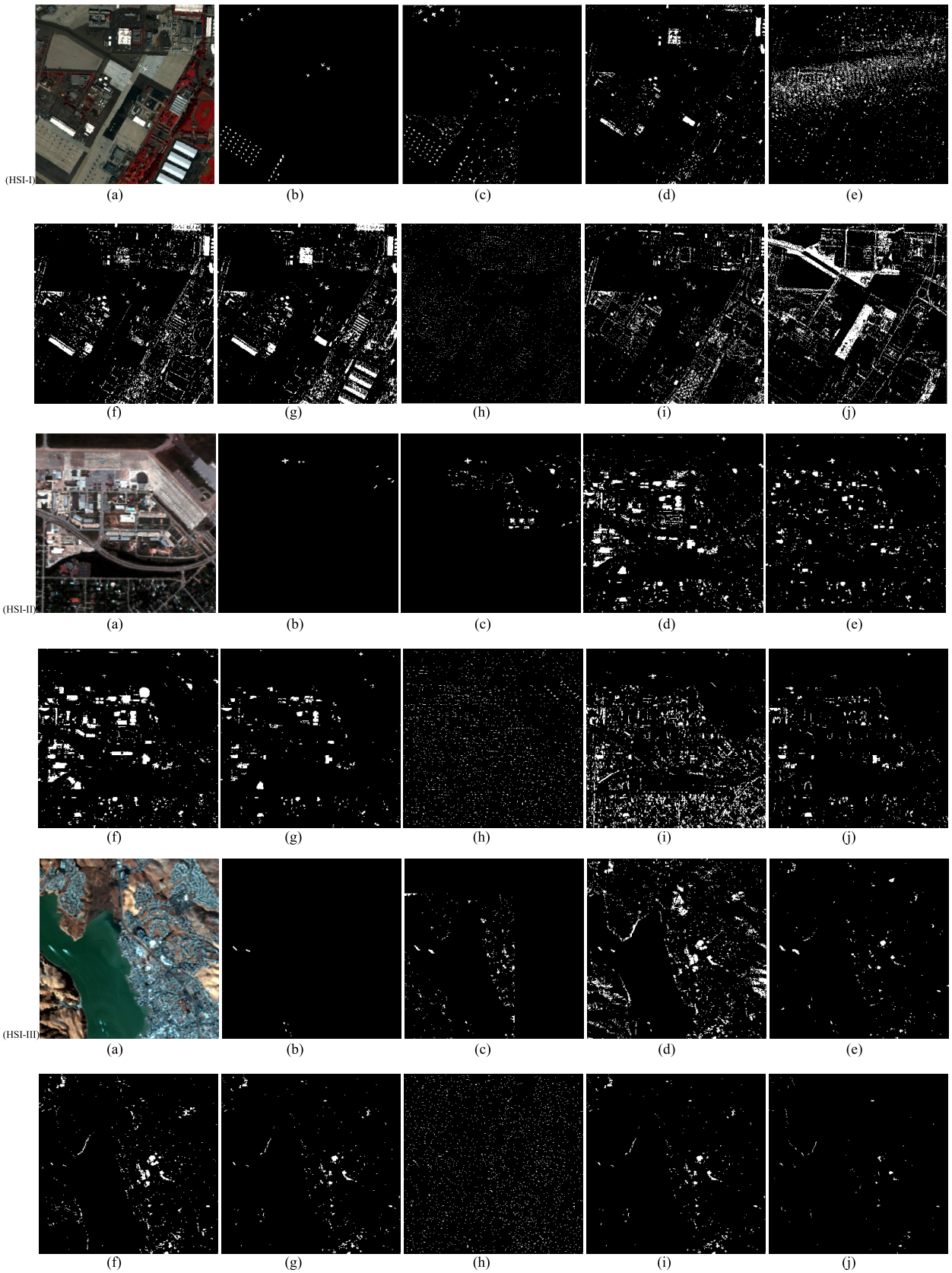


Fig. 17. Qualitative anomaly detection comparison of the compared methods on five HSI datasets: (a) input image; (b) ground truth; (c) proposed method; (d) GRX; (e) LRX; (f) LSDM-MoG; (g) RPCA-RX; (h) LSAD; (i) GTVLRR; and (j) LRaSMD.

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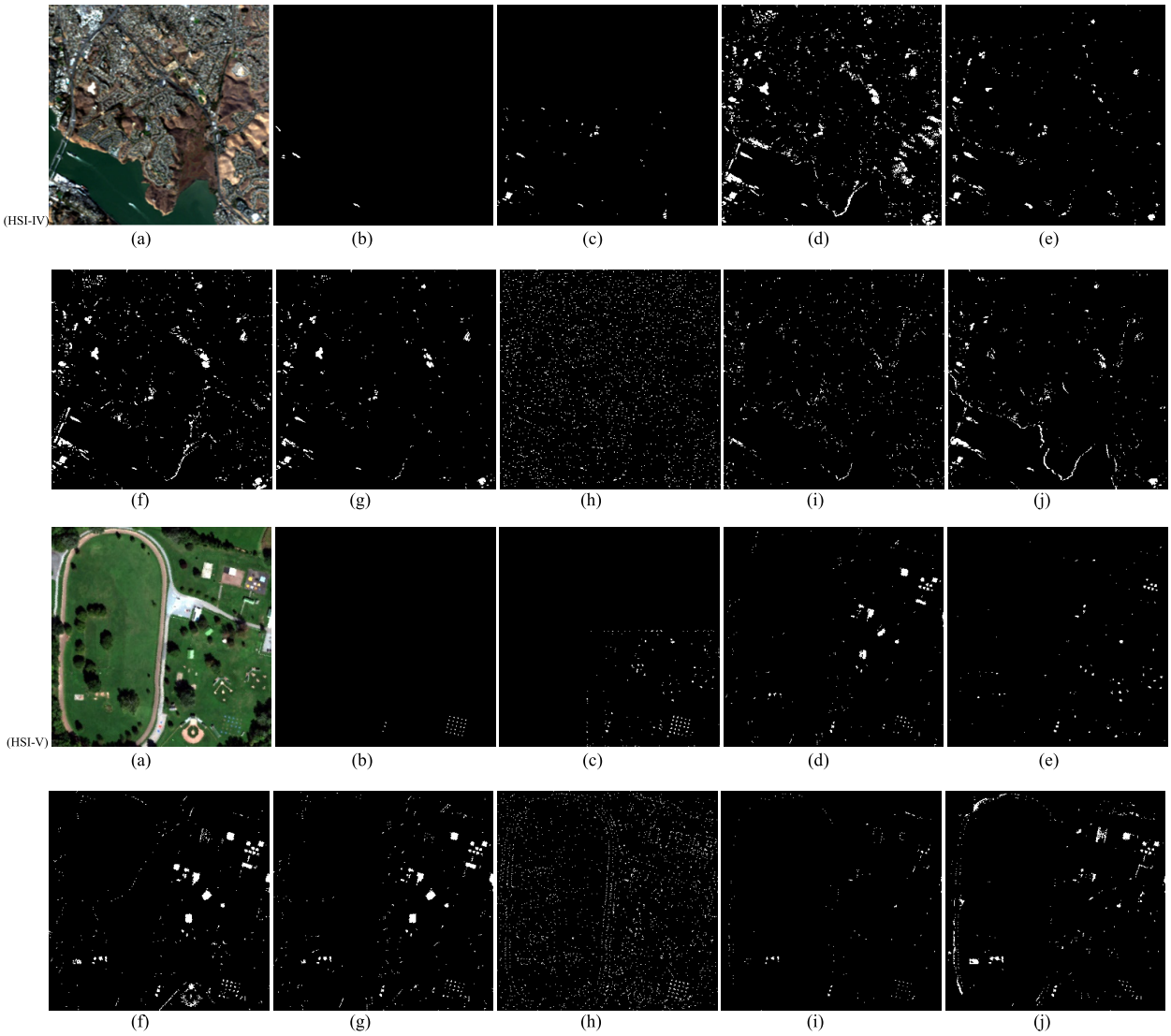


Fig. 17. *(Continued.)* Qualitative anomaly detection comparison of the compared methods on five HSI datasets: (a) input image; (b) ground truth; (c) proposed method; (d) GRX; (e) LRX; (f) LSDM-MoG; (g) RPCA-RX; (h) LSAD; (i) GTVLRR; and (j) LRaSMD.

images. The contrast-based method has more false alarms and a lower detection rate due to the abnormal detection of the whole image directly, while the proposed method can effectively suppress the background and highlight the targets, illustrating the advantages in the detection of complex scene images.

Furthermore, the ROC curve for each method is given in Fig. 18, indicating that the PRE-IS can provide higher detection accuracy but a lower false alarm rate than the GRX, LRX, LSDM-MoG, RPCA-RX, LSAD, GTVLRR, and LRaSMD methods. It proves the wide applicability of the PRE-IS to different anomaly detection applications. Table VI summarizes the AUC value of each detection method. As summarized in Table V, the proposed method outperforms the compared methods in terms of AUC, i.e., detection accuracy.

Table VII compares the false alarm rate of each method for the detection rate $= 0.9$. Under the same detection rate, the lower the false alarm rate indicates that the background suppression works better. The detection rate of 0.9 can be considered that the majority of anomaly targets are detected. Thus, in this article, the detection rate of 0.9 is used when the false alarm rate is compared for each method. A lower false alarm rate indicates a better background suppression in anomaly detection. The results prove that the proposed method can significantly decrease the false alarm by suppressing the background.

Table VIII compares the computation time (s) of the compared methods. All experiments were carried on a machine with an Intel Core i7-8700K CPU and a 32-GB RAM. Global methods usually provide less computation time than local ones, such as LRX and LSAD. Due to the complex regulation of the representation coefficient, GTVLRR and LSDM-MoG are of high computational complexity in detecting anomalies. The proposed method requires a certain amount of computations, especially in the tensor-RPCA with the total variation model involved in potential target region extraction. However, the proposed method utilizes the target region for anomaly detection, providing faster speed than direct anomaly detection that requires full image computation. The proposed method

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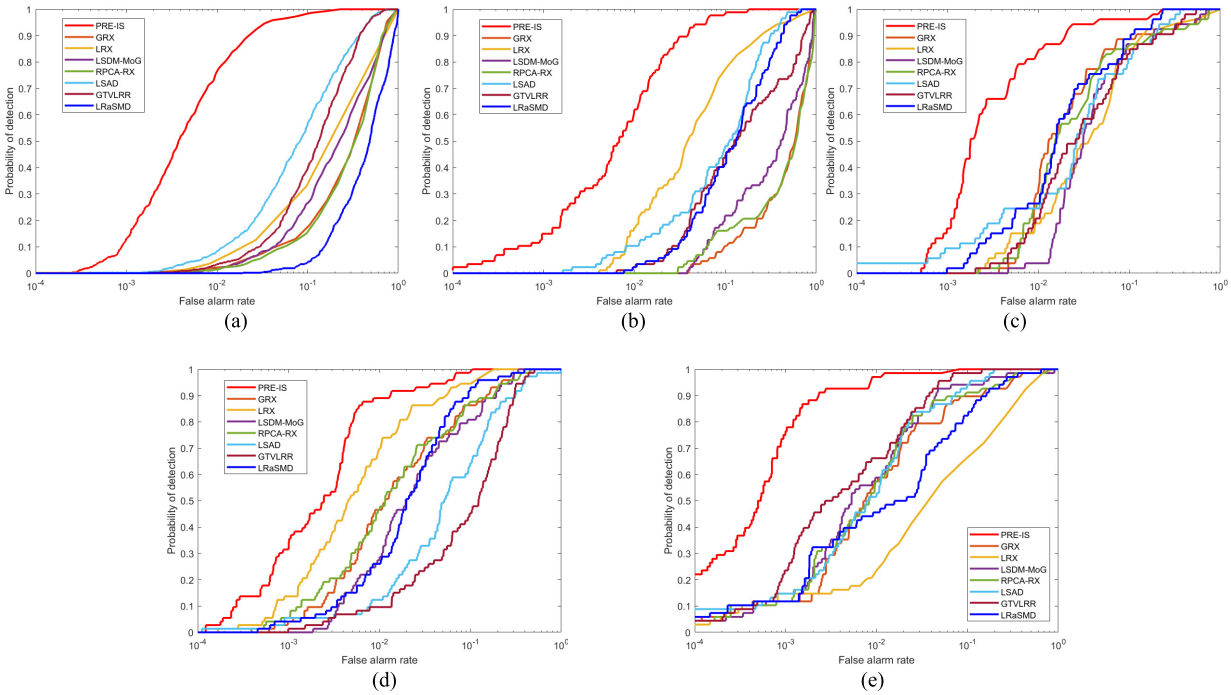


Fig. 18. ROC curves comparisons for different methods on: (a) HSI-I; (b) HSI-II; (c) HSI-III; (d) HSI-IV; and (e) HSI-V.

TABLE VI
AUC COMPARISON OF COMPARED METHODS

HSIs	PRE-IS	GRX	LRX	LSDM-MoG	RPCA-RX	LSAD	GTVLRR	LRaSMD
HSI-I	**0.9889**	0.6433	0.6180	0.6997	0.6376	0.8621	0.8262	0.5019
HSI-II	**0.9855**	0.5230	0.9038	0.5617	0.5432	0.8601	0.7339	0.8242
HSI-III	**0.9880**	0.9236	0.9197	0.9287	0.9236	0.9404	0.9496	0.9611
HSI-IV	**0.9927**	0.9536	0.9826	0.9441	0.9510	0.8890	0.8596	0.9600
HSI-V	**0.9979**	0.9537	0.8745	0.9667	0.9604	0.9769	0.9868	0.9442

TABLE VII
FALSE-ALARM RATES OF COMPARED METHODS

HSIs	PRE-IS	GRX	LRX	LSDM-MoG	RPCA-RX	LSAD	GTVLRR	LRaSMD
HSI-I	**0.023**	0.6545	0.6826	0.7029	0.6576	0.3782	0.3742	0.9032
HSI-II	**0.038**	0.9167	0.2589	0.910	0.9417	0.3424	0.7228	0.4116
HSI-III	**0.0197**	0.1208	0.1385	0.1836	0.1801	0.1932	0.1682	0.1034
HSI-IV	**0.013**	0.1532	0.055	0.1927	0.1827	0.306	0.301	0.094
HSI-V	**0.022**	0.1528	0.3895	0.0456	0.094	0.073	0.036	0.1619

TABLE VIII
COMPUTATION TIMES COMPARISON OF DIFFERENT METHODS WITH THE OPTIMAL WINDOW SIZE

HSIs	PRE-IS	GRX	LRX	LSDM-MoG	RPCA-RX	LSAD	GTVLRR	LRaSMD
HSI-I	457.86	0.58	321.47	172.76	26.86	3862.05	428.87	47.02
HSI-II	376.27	0.94	367.07	621.67	25.27	3241.42	270.38	49.41
HSI-III	365.98	0.62	435.23	198.35	24.78	3623.70	423.60	98.26
HSI-IV	378.66	0.86	392.36	245.36	23.72	7628.18	230.66	36.44
HSI-V	867.27	1.58	694.57	1112.32	99.25	9343.95	800.87	178.09

can achieve higher accuracy with a compromised computation time.

The anomaly background separability graph is used to further analyze the detection performance. Fig. 19 depicts the abnormal-background separability diagram of the compared methods. The red and blue boxes indicate the target and background, respectively, displaying the distribution of the anomaly and the background. The relative distance between the two boxes indicates the separability of the detection method. As shown in Fig. 19, the complexity of the background increases since a global image is used, inducing a challenge to distinguish between the target and the background. The proposed method can better separate the target pixel from the background pixel compared with the other methods.

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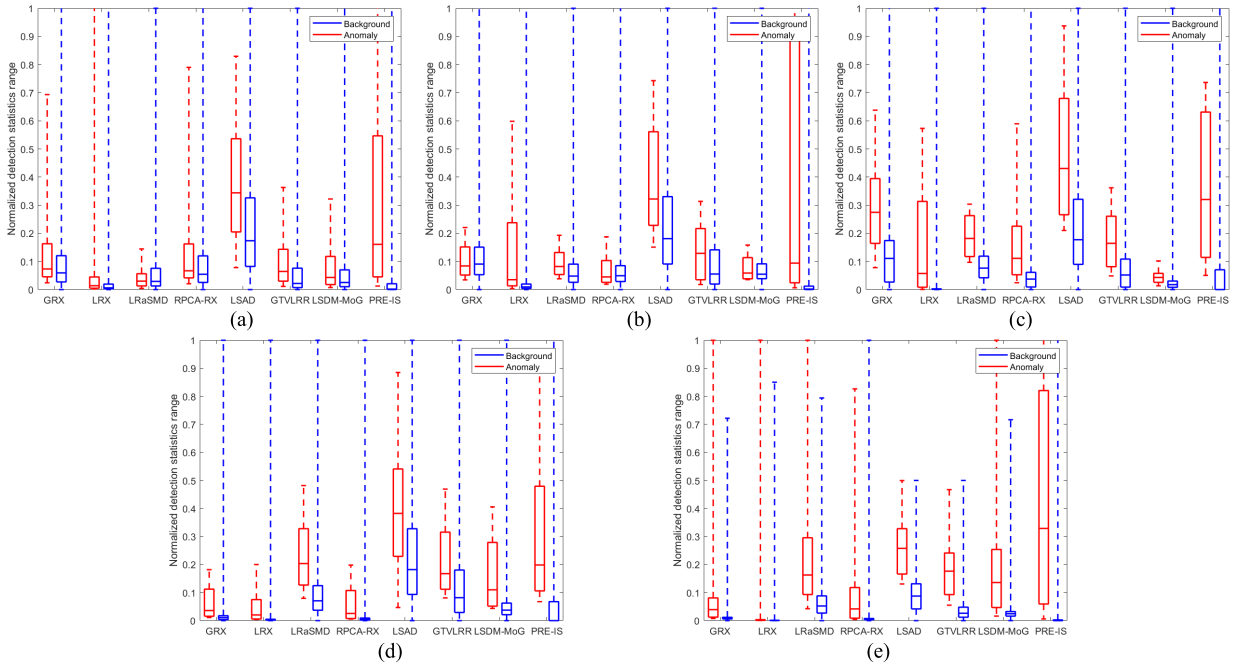


Fig. 19. Separability maps of detectors on five datasets: (a) HSI-I; (b) HSI-II; (c) HSI-III; (d) HSI-IV; and (e) HSI-V.

TABLE IX
INFLUENCE OF FOUR COMPARED METHODS

	HSI-I	HSI-II	HSI-III	HSI-IV	HSI-V
Proposed method	0.9889	0.9855	0.9880	0.9927	0.9979
Without Region Extraction	0.8639	0.9225	0.9588	0.9819	0.9843
Isolated forest of global mode	0.6305	0.8120	0.9469	0.9557	0.8922
SCR of local mode	0.8376	0.9138	0.9068	0.9697	0.9852

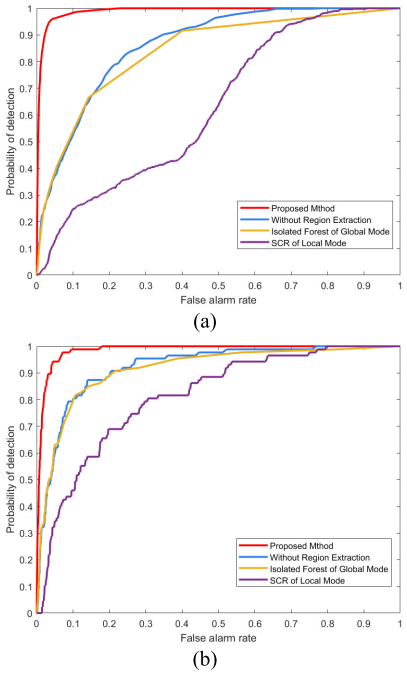


Fig. 20. ROC curves of the four compared methods in influence analysis on: (a) HSI-I and (b) HSI-II.

E. Ablation Study

This section conducts the ablation study to analyze the contribution of the potential target region extraction process to the entire detection performance. Similarly, the isolated forest of the global model and the SCR of the local model are compared in the task of anomaly detection. In this analysis, four methods with and without potential target region extraction, the global model, and the local model are compared in terms of AUC.

Table IX shows that the potential target region extraction process introduced to anomaly detection significantly induces an increased detection accuracy and a decreased false alarm rate. It shows that the potential target region extraction can accurately locate the subregions containing targets from the entire image scene through the low-rank sparse feature, thereby reducing the trouble in the subsequent detection process: false alarms caused by unrelated areas of the target. Combining the AUC value of each detection method in Table V, it can be seen that the AUC value of the abnormal detection part of the proposed method is higher than the AUC value of other compared methods on HSI-I and HSI-II. Also, the proposed method achieves comparable performance even without the region extraction on HSI-III, HSI-IV, and HSI-V. The maximum value of the AUC value of the comparison method in the comparison method is relatively close, indicating that the global to local anomaly detection method can more accurately find anomalous targets in a complex background, and the process of potential target regions removes false alarms. The isolated forest of the global model and the SCR of the local model are used to separately detect anomalies in the HSI. Table IX shows that there is indeed a defect. The combination of the global and local models can complement each other. As a result, the AUC value is enhanced, and the

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target is highlighted for detection, thereby achieving a better detection accuracy. In summary, it makes that the proposed method PRE-IS has obvious advantages for detecting anomalies in global HSIs with complex scenes, and the combination of global and local models can effectively highlight the target, which is conducive to the tracking of anomalous targets and improves the actual application value of anomaly detection.

A similar conclusion is drawn from the ROC curve in Fig. 20. As shown in the graph, the proposed method achieves a higher detection rate for the same false alarm rate. In contrast, the detection rate of the proposed method without region extraction, the isolated forest of the global model, and the statistical correlation of the local model are lower. The overall ROC curve of the proposed method is located left and above the curve of the other three methods. It shows that the potential target region extraction process improves the performance of the anomaly detection method, especially in false alarm reduction. The combination of global and local models can be effectively highlighted, reflecting the advantages of the proposed method for detecting anomalies in HSIs of complex scenes.

V. Conclusion

In this article, the PRE-IS is proposed for hyperspectral anomaly detection. The proposed model obtains the subregions that may contain the target through the tensor RPCA with full variation and the curvature circle model. The region traversing method is used to determine the potential target area. In order to enhance the sensitivity of different targets to the anomaly detector, we adopt an isolated forest of the global model and statistical correlation of the local model. The two results are combined through the adjustment factor for final anomaly detection. The experimental results on five public HSIs show that the proposed method can effectively detect the targets while reducing false alarms by potential target region extraction. Future works will include the consideration of the background complexity in the potential target region extraction process through the background dictionary construction.

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